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1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity four_to_sixteenDecoder is
5      port(input4_16_de: in std_logic_vector(3 downto 0);
6            output4_16_de: out std_logic_vector(15 downto 0));
7  end;
8
9  architecture synth of four_to_sixteenDecoder is
10 begin
11     output4_16_de(0) <= not input4_16_de(3) and not input4_16_de(2) and not
input4_16_de(1) and not input4_16_de(0);
12     output4_16_de(1) <= not input4_16_de(3) and not input4_16_de(2) and not
input4_16_de(1) and input4_16_de(0);
13     output4_16_de(2) <= not input4_16_de(3) and not input4_16_de(2) and input4_16_de(1)
and not input4_16_de(0);
14     output4_16_de(3) <= not input4_16_de(3) and not input4_16_de(2) and input4_16_de(1)
and input4_16_de(0);
15     output4_16_de(4) <= not input4_16_de(3) and input4_16_de(2) and not input4_16_de(1)
and not input4_16_de(0);
16     output4_16_de(5) <= not input4_16_de(3) and input4_16_de(2) and not input4_16_de(1)
and input4_16_de(0);
17     output4_16_de(6) <= not input4_16_de(3) and input4_16_de(2) and input4_16_de(1) and
not input4_16_de(0);
18     output4_16_de(7) <= not input4_16_de(3) and input4_16_de(2) and input4_16_de(1) and
input4_16_de(0);
19     output4_16_de(8) <= input4_16_de(3) and not input4_16_de(2) and not input4_16_de(1)
and not input4_16_de(0);
20     output4_16_de(9) <= input4_16_de(3) and not input4_16_de(2) and not input4_16_de(1)
and input4_16_de(0);
21     output4_16_de(10) <= input4_16_de(3) and not input4_16_de(2) and input4_16_de(1) and
not input4_16_de(0);
22     output4_16_de(11) <= input4_16_de(3) and not input4_16_de(2) and input4_16_de(1) and
input4_16_de(0);
23     output4_16_de(12) <= input4_16_de(3) and input4_16_de(2) and not input4_16_de(1) and
not input4_16_de(0);
24     output4_16_de(13) <= input4_16_de(3) and input4_16_de(2) and not input4_16_de(1) and
input4_16_de(0);
25     output4_16_de(14) <= input4_16_de(3) and input4_16_de(2) and input4_16_de(1) and not
input4_16_de(0);
26     output4_16_de(15) <= input4_16_de(3) and input4_16_de(2) and input4_16_de(1) and
input4_16_de(0);
27 end;
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